Video Streaming Overview

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Video Streaming is a core feature of GCXONE that enables real-time video delivery from integrated cameras and devices. The platform supports multiple streaming protocols and deployment models to provide optimal video quality and performance across different network conditions.

Key Capabilities

Multi-Protocol Support

- RTSP (Real-Time Streaming Protocol): Industry-standard protocol for IP cameras
- P2P (Peer-to-Peer): Direct streaming via Local Mode for reduced latency and enhanced performance
- HLS (HTTP Live Streaming): Adaptive streaming for compatible devices such as Camect
- TCP JPEG/Raw: Used for Milestone VMS integrations for optimal performance
- Manufacturer SDKs: Native protocols for Milestone, Avigilon, Hikvision (HikProConnect), and Dahua (DoLynk)
- WebSocket Streaming: Real-time streaming for web applications
- Low Bandwidth Mode: Optimized streaming for constrained network conditions

Streaming Modes

Cloud Streaming: Video routed through GCXONE cloud infrastructure. Accessible from anywhere with internet. Encrypted and secure with optimized adaptive quality.

Local Mode (P2P Streaming): Direct connection between operator workstation and device. Minimal latency for critical monitoring, reduced bandwidth on WAN, and enhanced audio quality via local SDK. Requires Local Mode service installation.

Stream Quality Management

GreenStream (Adaptive Streaming): GCXONE's intelligent streaming technology automatically selects the best matching video stream resolution based on the size of the viewing window on the operator's screen. This significantly decreases the load on the CPU and GPU while optimizing bandwidth usage.

Quality options:

- Auto (GreenStream): Automatically adjusts resolution based on viewport size and network conditions. Recommended.
- High: Maximum quality available for full-screen viewing
- Medium: Balanced quality and bandwidth for grid views
- Low: Optimized for limited bandwidth scenarios

GCXONE automatically adjusts stream quality based on viewport size, available network bandwidth, device capabilities, connection type, and number of concurrent streams.

How It Works

Cloud Streaming Architecture

1. Camera or NVR establishes connection to GCXONE platform
2. User requests live view from GCXONE interface
3. GCXONE requests video stream from device
4. Video is transcoded and optimized if needed
5. Stream delivered to user's browser or app with encryption

Local Mode Architecture

1. Local Mode service is installed on operator workstation
2. Service discovers devices on local network
3. Workstation establishes P2P connection to device
4. Video streamed directly, bypassing cloud
5. Automatically falls back to cloud if local connection fails

Supported Devices

- IP Cameras: Axis, Hikvision, Dahua, Hanwha, and more
- NVRs: All supported NVR manufacturers
- VMS Platforms: Milestone, Avigilon, Digital Watchdog, Genesis VMS
- Cloud Devices: NXG Cloud NVR, NXG Cloud Vision Edge
- Specialized Devices: Camect, Reconeyez, and others

Use Cases

Real-Time Monitoring

- Live surveillance of multiple locations
- Multi-camera grid views
- Mobile monitoring via mobile app
- Operator workstation monitoring

Critical Response

- Rapid access to live feeds during alarms
- Low-latency streaming for time-sensitive situations
- Multiple operators viewing the same camera simultaneously
- Priority streaming for emergency situations

Remote Access

- Access cameras from anywhere securely
- Mobile workforce monitoring
- After-hours surveillance

Deployment Considerations

Network Requirements

- Bandwidth: Minimum 2 Mbps per camera for standard quality
- Latency: Less than 100ms recommended for real-time monitoring
- Ports: RTSP (554), HTTP/HTTPS (80/443), and device-specific ports
- Firewall: Port forwarding may be required for direct access

Local Mode Requirements

- Workstation must be on the same network as devices
- Admin privileges required for Local Mode service installation
- Genesis Installer must be deployed
- Verify device supports Local Mode

Best Practices

Stream Quality Optimization

- Use Auto quality mode for best balance
- Enable Local Mode for high-camera-count sites
- Configure device bitrate settings appropriately
- Monitor bandwidth usage and adjust as needed

Performance Tips

- Limit concurrent high-quality streams per workstation
- Use grid views to monitor multiple cameras efficiently
- Close unused video windows to free resources
- Prefer Local Mode for operator workstations

Security

- Always use encrypted connections (HTTPS/RTSP over TLS)
- Implement proper authentication and authorization
- Use firewall rules to restrict access
- Monitor for unauthorized access attempts